

Unit - IV Sampling and Statistical Inference

In the present unit we discussing, Introduction to Sampling and Statistical inference, basics terminologies and definitions related to the sampling distributions, concepts of standard error and confidence interval, central limit theorem, Type - 1 and Type - 2 errors, Level of significance, one tailed and two tailed tests, Z-Test for single mean, for single proportion, for difference between means, Student t - test for single mean, for difference between two means, F - test for equality of two variances, Chi - square test for goodness of fit, for independence of attributes.

Continued Unit - IV on Introduction and Basics terminologies of Sampling Theory ...

Introduction:- In our day to day life it becomes often necessary to draw some valued and reasonable conclusions concerning a large mass of individuals or things known as population. It is difficult to take decisions regarding the statistical constants of population. Thus we prefer to examine a small part of this population known as sample with the motive of drawing valued conclusions about the statistical values of the entire population. This entire study of the behavior of the population through sample in a systematic way is called **Theory of Sampling**.

Example: Poll prediction general MP and MLA elections.

Statistics: Statistics deals with collection, classification and analysis of data and teaches how to make intelligent judgments, informed decision in the presence of uncertainty and variation.

Statistical Inferences: On the basis of sample information, we make certain decisions (conclusions) about the population, these decisions are said to be Statistical inferences.

Estimation: To estimate the parameters (Mean(μ), Variance(σ^2) and Standard deviation (σ) etc.), we use the value of sample statistics (sample mean($\bar{x}$), sample variance (s^2) and sample standard deviation(s)). Thus a statistic is an estimate of parameter and this process is said to be **Statistical estimation**.

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There are two types of estimations, they are,

Point estimate: A point estimate of a parameter is a number (a point on the real line), which is computed from the given sample and serve as an approximation of the unknown exact value of the parameter. The objective of a point estimation is to use a sample to compute a number that represent in some sense or good guess for the true value of the parameter.

Interval estimation: An interval estimate is an interval (confidence interval) obtained from the sample, in which population parameter may be expected to lie with a given degree of confidence. A parameter is usually estimated to be within a range of values, rather than a single number, is an interval estimate and the process of estimating with a range of values is known as interval estimation.

Hypothesis: An assumption made about the population parameter, which may be true or false, is called **Hypothesis**.

Testing of Hypothesis : A process to decide whether to accept or reject hypothesis is called testing of hypothesis. Tests of hypothesis are also called **Tests of Significance**.

Notations of parameters and statistic:

Population	Sample
Population size = N	Sample number = n
Population mean = μ	Sample mean = $\bar{X}$ (OR) $\mu_{\bar{X}}$
Population variance = σ^2	Sample Variance = s^2 (OR) σ_s^2
Population S.D. = σ	Sample S. D. = s (OR) σ_s

Continued Unit - IV on Basics terminologies of Sampling Theory ...

Sampling Theory: Sampling theory is a study of relationship existing between Population and sample drawn from Population.

Main objectives of Sampling:

The following are main objectives of the Sampling

- 1 Aims at gathering the maximum information about the Population with minimum effort, cost and time.
- 2 To obtain the best possible values of the parameters under specific conditions.
- 3 Determine reliability of these estimates.

Some important terms and definitions

Population: It is a set of all aggregate objects under study. A finite or infinite collection or aggregate of items under consideration. Also refers to the totality of the observation or any collection of individuals or results of an operation that can be numerically specified.

Population size: The number of objects or observations in Population is called Population size. The population size is denoted by N . The Population is said to be finite or infinite depending on the Population size N being finite or infinite.

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Sample: It is a subset of the Population and are considered to study the behavior of the Population. A finite subset of Population or small selection selected from the population is called a sample.

In other words, for estimating the population characteristics, a part of the population will be studied, that part is called a sample.

Sample size: The number of objects or observations in the sample is called a sample size. The sample size is denoted by n .

Note:

- 1 If sample size $n \geq 30$, then the is considered to be large sample.
- 2 If sample size $n < 30$, then the sample is considered to be small sample.

Parameters: The statistical constants of the population such as population mean (μ), population variance (σ^2), population standard deviation(σ) etc., are called parameters.

Statistic: The statistical constants of the sample such as sample mean($\bar{x}$), sample variance (σ^2), sample standard deviation (σ) etc., are called statistic.

Sampling: The process of drawing (or) selecting sample's from the population is called **sampling**.

Random Sampling: A sampling in which each member of the population has equal chance (or) probability of being included in to the sample is called **random**

Sampling.

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Random sample: A sample obtained through the random sample is called random sample.

Simple Sampling: Random sampling in which each unit of population has an equal chance 'p' of being included in the sample and that the probability is independent of the previous drawings.

Note: There are two types of sampling (i) Sampling with replacement.

(ii) Sampling without replacement.

- ① Hence for simple sampling either the population must be infinite (or) if it is finite then drawing must be made with replacement.
- ② There are NC_n different samples of size n that can be picked up from a population of size N in sampling without replacement where as N^n possible samples with replacement.

Sampling with replacement: If the items are selected one by one in such a way that an item drawn at a time is replaced back to the population before the next (or) subsequent draw, such a sampling is known as **sampling with replacement**. In this type of sampling from a population of size N , the probability of selection of a unit at each draw remains $\frac{1}{N}$.

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Sampling without replacement: A sampling in which an item of the population cannot be chosen more than once is known as **Sampling without replacement**.

Sample mean: If $x_1, x_2, x_3, \dots, x_n$, constitute a random sample of size n from a population then

❶ sample mean ($\bar{x}$):-

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}.$$

❷ Sample Variance (s^2):-

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}.$$

❸ **Sample standard deviation (s)**:- s , is the positive square root of the variance.

Sampling Distribution:- The probability distribution associated with any statistic is called the **sampling distribution** of that statistic. It describes how a statistic will vary from one sample to other sample of the same size. For instance, consider all possible samples of size n which can be drawn from a given population. For each sample we can calculate the mean (or variance or proportion etc). The value of the mean will not be identical from sample to sample.

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If we group all these means according to their frequencies, the frequency distribution so formed is known as sampling distribution of means.

Standard Error (S. E.):- The standard error is the standard deviation of the sampling distribution. For assessing the difference between the expected value and observed value, the standard error is used (The reciprocal of standard error is called as Precision). That is S.E. is a measure of variability of the statistic and is useful in estimation and testing of hypothesis.

- 1 In the theory of estimation S.E. is used to decide the efficiency and consistency of the statistic as an estimator.
- 2 In interval estimation, standard error is used to write down the confidence intervals.
- 3 In testing of hypothesis, standard error of the test statistic is used to standardize the distribution of the statistic.

Sampling distribution of Means (σ , known)

Let the population be infinitely large having a population mean μ and standard deviation σ . If a random sample of size n is taken from an infinite population then,

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1

$$E(\bar{x}) = \mu_{\bar{x}} = \mu.$$

2 that is, (the expected value of the sample means) = (population mean).

3

$$\sigma_{\bar{x}}^2 = V(\bar{x}) = \frac{\sigma^2}{n}.$$

4 (the variance of sample means) = (the variance of population mean divided by the sample size).

5 Therefore, the standard deviation of $\bar{x}$ is,

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}.$$

Sampling with replacement:- When sampling is done with replacement, so that the population is back to the same form before the next sample member is picked up. Therefore, we have,

1

$$E(\bar{x}) = \mu_{\bar{x}} = \mu.$$

2

$$\sigma_{\bar{x}}^2 = V(\bar{x}) = \frac{\sigma^2}{n}.$$

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1

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}.$$

Sampling without replacement:- When sample is picked up without replacement from a finite population, the probability distribution of second random variable depends on the outcome of the first pick up. So n sample members don't remain independent, therefore, we have,

1

$$E(\bar{x}) = \mu_{\bar{x}} = \mu.$$

2

$$\sigma_{\bar{x}}^2 = V(\bar{x}) = \frac{\sigma^2}{n} \left(\frac{N-n}{N-1} \right).$$

3

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} = \frac{\sigma}{\sqrt{n}}.$$

(Approx.) (if $\frac{n}{N}$ is very small)

Note:

1 $\frac{N-n}{N-1}$ is known as finite population corrector factor.

2 The factor $\frac{N-n}{N-1}$ can be neglected if N is too large compared to the sample size n .

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Sampling from a Normal population:

If $X \approx N(\mu, \sigma^2)$ then it follows that $\bar{X} \approx N(\mu, \frac{\sigma^2}{n})$

Problem - 01: A population consists of five numbers 2, 3, 6, 8 and 11. Consider all possible samples of size 2 that can be drawn with replacement and without replacement from the population. Find (i) The mean of the population (ii) the standard deviation of the population (iii) the mean of the sampling distribution of means (iv) the standard deviation (S.E.) of the sampling distribution of means.

Solution: (i) Mean of the population =

$$\mu = \frac{\sum_{i=1}^5}{N} = \frac{2 + 3 + 6 + 8 + 11}{5} = 6.$$

(ii) Variance of population =

$$\sigma^2 = \frac{\sum_{i=1}^5 (x_i - \mu)^2}{N} = \frac{(2 - 6)^2 + (3 - 6)^2 + (6 - 6)^2 + (8 - 6)^2 + (11 - 6)^2}{5} = 10.8.$$

Therefore, standard deviation (S.E.)=

$$\sigma = \sqrt{\text{Variance}} = \sqrt{10.8} = 3.29.$$

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(a) Samples with replacement:

(iii) The total number of samples with replacement: is $N^n = 5^2 = 25$ of sample size $n = 2$.

$(2, 2), (2, 3), (2, 6), (2, 8), (2, 11), (3, 2), (3, 3), (3, 6), (3, 8), (3, 11),$
 $(6, 2), (6, 3), (6, 6), (6, 8), (6, 11),$
 $(8, 2), (8, 3), (8, 6), (8, 8), (8, 11),$
 $(11, 2), (11, 3), (11, 6), (11, 8), (11, 11).$

corresponding sample means are:

2, 2.5, 4, 5, 6.5,
2.5, 3, 4.5, 5.5, 7,
4, 4.5, 6, 7, 8.5,
5, 5.5, 7, 8, 9.5,
6.5, 7, 8.5, 9.5, 11.

the mean of the sampling distribution of means

$$\mu_{\bar{x}} = \frac{2 + 2.5 + 4 + 5 + 6.5 + 2.5 + 3 + 4.5 + 5.5 + 7 + 4 + 4.5 + 6 + 7 + 8.5 + 5 + 5.5 + 7}{25}$$

Therefore, $\mu_{\bar{x}} = \mu$

Hence, The population mean (μ) is equal to the mean of the sampling distribution of means ($\mu_{\bar{x}}$).

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(iv) The variance of the sampling distribution of means,

$$\sigma_{\bar{x}}^2 = \frac{\sum_{i=1}^{25} (\bar{x}_i - \mu_{\bar{x}})^2}{25}$$

$$\sigma_{\bar{x}}^2 = \frac{(2 - 6)^2 + (2.5 - 6)^2 + (4 - 6)^2 + \dots + (11 - 6)^2}{25}$$

$$\sigma_{\bar{x}}^2 = 5.4$$

Hence, $\frac{\sigma^2}{n} = \frac{10.8}{2} = 5.4$

Therefore, $\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}$.

(b) Samples without replacement:

(i) Population mean = $\mu = 6$ and

(ii) Population Variance = $\sigma^2 = 10.8$, Population standard deviation = $\sigma = 3.29$.

(iii) **The total number of samples without replacement:** is

$${}^N C_n = {}^5 C_2 = \frac{(5!)}{(2!)(5-2)!} = 10$$

of samples of size $n = 2$.

$(2, 3), (2, 6), (2, 8), (2, 11), (3, 6), (3, 8), (3, 11), (6, 8), (6, 11), (8, 11)$

and the corresponding sample means are:

2.5, 4, 5, 6.5, 3, 5.5, 7, 7, 8.5, 9.5

Therefore, the mean of the sampling distribution of means

$$\mu_{\bar{x}} = \frac{2.5 + 4 + 5 + 6.5 + 3 + 5.5 + 7 + 7 + 8.5 + 9.5}{10} = 6$$

Therefore,

the population mean(μ)= the mean of the sampling distribution of means (μ).

(iv) The variance of the sampling distribution of means,

$$\sigma_{\bar{x}}^2 = \frac{\sum_{i=1}^{10} (\bar{x}_i - \mu_{\bar{x}})^2}{10}$$

$$\sigma_{\bar{x}}^2 = \frac{\sum_{i=1}^{10} (\bar{x}_i - \mu_{\bar{x}})^2}{10}$$

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$$\sigma_{\bar{x}}^2 = \frac{(2.5 - 6)^2 + (4 - 6)^2 + (5 - 6)^2 + (6.5 - 6)^2 + (4.5 - 6)^2 + (5.5 - 6)^2 + (7 - 6)^2 + (6 - 6)^2}{10}$$

$$\sigma_{\bar{x}}^2 = 4.05$$

Therefore, the variance of the sampling distribution of means $= \sigma_{\bar{x}}^2 = 4.05$.

Hence, the standard deviation (S.E.) of the sampling distribution of mean $= \sigma_{\bar{x}} = 2.01$

Therefore, it shows that,

$$\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n} \left(\frac{N - n}{N - 1} \right) = \frac{10.8}{2} \left(\frac{5 - 2}{5 - 1} \right) = 4.05$$

Problem - 02: A population consists of four numbers 3, 7, 11 and 15. Consider all possible samples of size 2 that can be drawn with replacement and without replacement from the population. Find (i) The mean of the population (ii) the standard deviation of the population (iii) the mean of the sampling distribution of means (iv) the standard deviation (S.E.) of the sampling distribution of means.

Solution:

(i) the mean of the population is

$$\mu = \frac{\sum_{i=1}^4}{N} = \frac{3 + 7 + 11 + 15}{4} = 9$$

(ii) The variance of population is

$$\sigma^2 = \frac{\sum_{i=1}^4 (x_i - \mu)^2}{N} = \frac{(3 - 9)^2 + (7 - 9)^2 + (11 - 9)^2 + (15 - 9)^2}{4} = 20$$

Therefore, the standard deviation (S.E.) is

$$\sigma = \sqrt{\text{Variance}} = \sqrt{20} = 4.472$$

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(a) Samples with replacement:

(iii) **The total number of samples with replacement:** is $N^n = 4^2 = 16$ of sample size $n = 2$.

(3, 3), (3, 7), (3, 11), (3, 15),

(7, 3), (7, 7), (7, 11), (7, 15),

(11, 3), (11, 7), (11, 11), (11, 15),

(15, 3), (15, 7), (15, 11), (15, 15),

corresponding sample means are:

3, 5, 7, 9,,

5, 7, 9, 11,

7, 9, 11, 13,

9, 11, 13, 15,

the mean of the sampling distribution of means is

$$\mu_{\bar{x}} = \frac{3 + 5 + 7 + 9 + 5 + 7 + 9 + 11 + 7 + 9 + 11 + 13 + 9 + 11 + 13 + 15}{16} = 9$$

= .

Therefore, $\mu_{\bar{x}} = \mu$

Therefore,

The population mean(μ)= the mean of the sampling distribution of means (μ).

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(iv) The variance of the sampling distribution of means,

$$\sigma_{\bar{x}}^2 = \frac{\sum_{i=1}^{16} (\bar{x}_i - \mu_{\bar{x}})^2}{n}$$

$$\sigma_{\bar{x}}^2 = \frac{(3-9)^2 + (5-9)^2 + (7-9)^2 + \dots + (15-9)^2}{16}$$

$$\sigma_{\bar{x}}^2 = 10$$

The standard deviation (S.E.) of the sampling distribution of means $\sigma_{\bar{x}} = 3.16$.

$$\text{Hence, } \frac{\sigma^2}{n} = \frac{20}{2} = 10$$

$$\text{Therefore, } \sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}.$$

(b) Samples without replacement:

(i) Population mean $= \mu = 9$ and (ii) Population Variance $= \sigma^2 = 20$, Population standard deviation $= \sigma = 4.472$.

(iii) **The total number of samples without replacement:** is

$$NC_n = 4C_2 = \frac{(4!)}{(2!(4-2)!)} = 6, \text{ of samples of size } n = 2.$$

(3, 7), (3, 11), (3, 15), (7, 11), (7, 15), (11, 15)

and the corresponding sample means are: 5, 7, 9, 9, 11, 13,

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Therefore, the mean of the sampling distribution of means

$$\mu_{\bar{x}} = \frac{5+7+9+9+11+13}{6} = 9$$

Therefore,

The population mean(μ)= the mean of the sampling distribution of means (μ).

Therefore, the variance of the sampling distribution of means =

$$\sigma_{\bar{x}}^2 = \frac{(5-9)^2 + (7-9)^2 + (9-9)^2 + (9-9)^2 + (11-9)^2 + (13-9)^2}{6} = \frac{20}{3}.$$

Hence, the standard deviation (S.E.) of the sampling distribution of mean =

$$\sigma_{\bar{x}} = 2.58$$

Therefore, it shows that,

$$\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n} \left(\frac{N-n}{N-1} \right) = \frac{20}{2} \left(\frac{4-2}{4-1} \right) = \frac{20}{3}.$$

Problem - 03: A population consists of six numbers 4, 8, 12, 16, 20 and 24. Consider all possible samples of size 2 that can be drawn with replacement from the population. Find (i) The mean of the population (ii) the standard deviation of the population (iii) the mean of the sampling distribution of means (iv) the standard deviation (S.E.) of the sampling distribution of means.**Home work**

Continued Unit - IV on Sampling Theory ...

Theorem - 01:- The expected value of the sample means ($\mu_{\bar{x}}$) is equal to the population mean (μ).

$$\mu_{\bar{x}} = \mu \quad (1)$$

Theorem - 02:- If a population is infinite and the sampling is random (or) if the population is finite and sampling is with replacement the variance of the sampling distribution of means, denoted by $\sigma_{\bar{x}}^2$, is given by

$$\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n} \quad (2)$$

where σ^2 is the population variance and n is sample size.

Theorem - 03:- If a population is finite of size N and if sampling is without replacement then

$$\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n} \left(\frac{N-n}{N-1} \right) \quad (3)$$

$\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}$, approx., when N is very large i.e., as $N \rightarrow \infty$.

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Theorem - 04:- If the population from which samples are taken is normally distributed with mean (μ) and variance (σ^2), then the sample mean is normally distributed with mean (μ) and variance $\frac{\sigma^2}{n}$.

Theorem - 05:- (Central limit theorem)

Suppose that the population from which samples are taken has a probability distribution with mean (μ) and variance (σ^2) that is not necessarily a normal distribution. Then the standardized variable associated with ($\bar{x}$) is given by

$$z = \frac{(\bar{x} - \mu)}{\frac{\sigma}{\sqrt{n}}} \quad (4)$$

is asymptotically normal, that is

$$\lim_{n \rightarrow \infty} P(Z \leq z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-\frac{z^2}{2}} dz \quad (5)$$

Note: It is assumed here that the population is infinite (or) that sampling is with replacement. Otherwise the above is true $\frac{\sigma}{\sqrt{n}}$ in equation (4) by $\sigma_{\bar{x}}$ from equation (3).

Conclusion: If the population is normal, then the sampling distribution of mean is also normal with mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$.

While for large samples ($n \geq 30$) the same result hold even if the distribution of the population is non - normal.

Hypothesis Testing:-

Suppose we assume a certain value to the unknown population parameter and also estimated using sample values. If the sample value is exactly same (or) very close to our assumption then it can be accepted as the parameter. Unlike, if it is far away, then we reject it. But if it is neither close nor far away, then it is require to follow a procedure to decide whether to accept the presumed value (or) not, on the basis of sample values, such a procedure is called **Testing of hypothesis**.

Hypothesis:- An assumption made about the population parameter, which may be true (or) false, is called **hypothesis**.

Testing of hypothesis:- A process to decide whether to accept (or) reject hypothesis is called **Testing of hypothesis**. Tests of hypothesis are also called **Tests of Significance**.

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Testing of hypothesis is classified in to two types, they are

- 1 Null Hypothesis.
- 2 Alternative hypothesis.

Null hypothesis:- The hypothesis formulated for the sake of rejection, under the assumption that it is true is called **Null hypothesis** and it is denoted by H_0 .

Alternative hypothesis:- The hypothesis which is contradictory to the null hypothesis is called an **Alternative hypothesis** and it is denoted by H_1 (or) H_a .

Note:

- 1 The null hypothesis will be rejected in favour of the alternative hypothesis only when sample evidence suggests that H_0 is false.
- 2 The two possible conclusions from a hypothesis testing analysis are reject H_0 (or) fail to reject H_0

Error's in Hypothesis testing:-

The following two types of errors are likely to happen while testing of hypothesis.

Type - I Error: Rejecting the null hypothesis when it is true is called **type - I error**.

Type - II Error: Not rejecting the null hypothesis when it is false is called **type - II error**.

	Accept	Reject
$H_0 : \text{True}$	Correct decision	Type - I error
$H_0 : \text{False}$	Type - II error	Correct decision

One tailed and Two tailed Tests:

One tailed test:- A test of any statistical hypothesis, where the alternative hypothesis is one - sided such as if $H_0 : \mu = \mu_0$ then

$H_1 : \mu > \mu_0$ (Right tailed (or) upper tailed) (OR)

$H_1 : \mu < \mu_0$ (left tailed (or) lower tailed) - is called **One tailed test**.

Two tailed test:- A test of any statistical hypothesis where the alternative hypothesis is two - sided such as, if $H_0 : \mu = \mu_0$ then $H_1 : \mu \neq \mu_0$ is called a **Two tailed test**.

Note: The null hypothesis will always be stated using equality sign so as to specify a single value. In this way the probability of committing a Type I error can be controlled.

Level of Significance:- The probability of type - I error (Rejecting H_0 when it is true) below which we reject the null hypothesis is known as **Level of Significance(LOS)**.

In other words, the maximum value of type - I error which we would be willing to risk is called **Level of Significance(LOS)**.

Note:

- ➊ In practice there are two types such as (0.05) and (0.01) are commonly accepted values of the **Level of Significance(LOS)**.
- ➋ If Level of Significance(LOS) is (0.05), it simply means that on an average in 5 chances out of 100 we are likely to reject a correct H_0 and we would be 95% confident of our decision while rejecting a null hypothesis.

Critical Region:- The region in which the sample value falls and null hypothesis will be rejected is known as **critical region** (or) **region of rejection**.

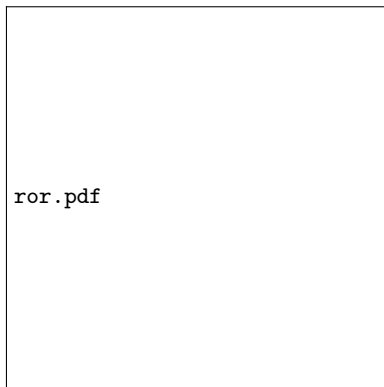


Figure 1: Region of Rejection (α)

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Significant value (OR) Critical value:- The value of a test statistic which separates the critical region and the region of acceptance is called the **Significant value (or) critical value**. It depends totally on

(i) level of significance, (ii) The alternative hypothesis.

Critical values for 1 % and 5 % Level of significance

Tests	1% level	5% level
Two tailed	$ z_{\frac{\alpha}{2}} = 2.58$	$ z_{\frac{\alpha}{2}} = 1.96$
Right tailed	$ z_{\alpha} = 2.33$	$ z_{\alpha} = 1.645$
Left tailed	$ z_{\alpha} = -2.33$	$ z_{\alpha} = -1.645$

Power of a Test: The power of a test is the ability of a statistical test to detect the alternative hypothesis when it is true. The power of a test is $1 - \beta$

In otherwords, if the probability of accepting a false hypothesis is β then the power of a test is $1 - \beta$.

Note:

❶ $P[\text{Rejecting } H_0 \text{ when it is true}] = P[\text{type I error}] = \alpha.$

❷ $P[\text{Accepting } H_0 \text{ when it is false}] = P[\text{type II error}] = \beta.$

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Note - 01: Normally we use two - tailed test. If we have to apply one - tailed test, then for the level of significance of $\alpha\%$. We have to refer to $2\alpha\%$ in two - tailed table.

Note - 02: The following tests are considered for the estimation of the parameter of the population.

z - test:

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Here, $\bar{x}$ is the sample mean, μ be the population mean, σ^2 be the population variance and n be the sample size. This test statistic is used when the sample size is $n > 30$ (for large samples) and the standard deviation of the population σ is known.

Student's t - test:

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

Here

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

where $(n - 1)$ are degrees of freedom and here, $\bar{x}$ is the sample mean, μ be the population mean, s^2 be the sample variance and n be the sample size. This test statistic is used when the sample size is $n < 30$ (for small samples).

Note - 01: In the problems, we apply the formulae and estimate the calculate the test value, that is t_{cal} . If the calculated value is less than the tabulated value, our hypothesis is accepted at the desired level of significance. If the calculated value is greater than the tabulated value then we reject the hypothesis.

Note - 02: Applying the z - test, for the large samples, that is, when the sample size $n > 30$ and the standard deviation of the population σ is known.

Note - 03: Applying the student's t - test, for the small samples, that is, when the sample size $n < 30$.

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Confidence limits OR confidence intervals:

It is a type of estimate computed from the statistics of the observed data. This proposes a range of possible values for an unknown parameter. There are two types of confidence intervals, they are

Formulas for the z - test:

Formula - 01: 95% OR 5% confidence interval:

In the 95% confidence interval the sample statistic can be expected to lie in the below interval for 95% times $\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}}$.

Formula - 02: 99% OR 1% confidence interval:

In the 99% confidence interval the sample statistic can be expected to lie in the below interval for 99% times $\bar{x} \pm 2.58 \frac{\sigma}{\sqrt{n}}$.

Formulas for the student's t - test:

We have to refer the table value for a given degrees of freedom(d. f.)

Formula - 01: 5% confidence interval:

In the 5% confidence interval the sample statistic can be expected to lie in the below interval for 5% times $\bar{x} \pm \frac{s}{\sqrt{n}} t_{0.05}$. (at a given d. f. at 5%), Here s^2 is the sample variance.

Formula - 02: 1% confidence interval:

In the 1% confidence interval the sample statistic can be expected to lie in the below interval for 1% times $\bar{x} \pm \frac{s}{\sqrt{n}} t_{0.01}$. (at a given d. f. at 1%), Here s^2 is the sample variance.

Continued Unit - IV problems on Sampling Theory ...

Problem - 01: The nine items of a sample have the following values 45, 47, 50, 52, 48, 47, 49, 53, 51. Does the mean of these differ significantly from the assumed mean of the population 47.5 at 5% and 1% level of significance. Also find the 95% and 99% confidence interval for the mean.

Solution: Given, the sample size $n = 9$

The mean of the sample $= \bar{x} = \frac{\sum x}{n} = \frac{45+47+\dots+51}{9} = 49.11$

sample variance $= s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$

$$s^2 = \frac{(45 - 49.11)^2 + (47 - 49.11)^2 + \dots + (51 - 49.11)^2}{9 - 1} = 6.86$$

sample S. D. $= s = \sqrt{6.86} = 2.61$

We shall frame the hypothesis

Null hypothesis : $H_0 : \mu = 47.5$

Alternative hypothesis : $H_1 : \mu \neq 47.5$

We shall find test statistic

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{49.11 - 47.5}{\frac{2.61}{\sqrt{9}}} = 1.83$$

$\therefore t_{cal} = 1.83$

Continued Unit - IV problems on Sampling Theory ...

The tabulated value at 5% LOS: From the table $t_{0.05}$ with $n - 1 = 8$ d. f. = 2.306, $t_{tab} = 2.306$. Therefore, the $t_{cal} < t_{tab}$ and we can accept the null hypothesis $H_0 = 47.5$ at 5% LOS.

The tabulated value at 1% LOS: From the table $t_{0.01}$ with $n - 1 = 8$ d. f. = 3.355, $t_{tab} = 3.355$. Therefore, the $t_{cal} < t_{tab}$ and we can accept the null hypothesis $H_0 = 47.5$ at 1% LOS.

Confidence intervals:

In the 95% confidence intervals is $\bar{x} \pm \frac{s}{\sqrt{n}} t_{0.05} = 49.11 \pm 2.306(\frac{2.61}{\sqrt{9}}) = (47.10, 51.11)$. Hence the sample statistic can be expected to lie in the interval (47.10, 51.11) for 95% of times.

In the 99% confidence intervals is $\bar{x} \pm \frac{s}{\sqrt{n}} t_{0.01} = 49.11 \pm 3.355(\frac{2.61}{\sqrt{9}}) = (46.19, 52.02)$. Hence the sample statistic can be expected to lie in the interval (46.19, 52.02) for 99% of times.

Note: If the null hypothesis H_0 is true at 5% LOS, it is also true at 1% LOS.

Problem - 02: A random sample of 10 boys had the following I.Q.'s 70, 120, 110, 101, 88, 83, 95, 107, 100. Do these data support assumption of population I. Q.'s is equal to 100 at 1% and 5% LOS. Also find 99% and 95% confidence intervals for the mean.

Tests of Significance

I Large Samples:- There are four important tests available for testing the significance of large samples.

- ① Test of significance of single mean.
- ② Test of significance for difference of means.
- ③ Test of significance for difference of standard deviations.
- ④ Test of significance for difference of proportions.

I Small Samples:-

(i) **Student's t - distribution (OR) Student's t - Test:-**

- ① Test of significance of single mean (σ unknown) (whether sample mean deviates significantly from proportion mean).
- ② Test the significance for difference between two means (of independent samples).
- ③ Test of significance for difference between two means (of dependent samples or paired observations).
- ④ Test of significance of an observed correlation or regression coefficients etc.

Continued Unit - IV on Sampling Theory ...

(ii) **F - Distribution (OR) F - Test:** To test the significance of difference between variances.

(iii) **χ^2 - Distribution (OR) χ^2 - Test (Chi - squared distribution)**

- ➊ To test the goodness of fit.
- ➋ To test the independent of attributes.
- ➌ To test the homogeneity of independent estimates of the population variance.
- ➍ To combine various probability obtained from independent experiments to give a single set of significance.
- ➎ To test the homogeneity of independent estimates of the population correlation coefficient.
- ➏ To test if the hypothetical value of the population variance is $\sigma^2 = \sigma_0^2$. etc.

Working procedure for testing of hypothesis:

Step - 1: Set up the null hypothesis.

Step - 2: Set up the alternative hypothesis.

Step - 3: Identify the test statistic.

Step - 4: Set Level of Significance as 1% (or) 5% .

Step - 5: Identify the critical region.

Step - 6: Make decision based on calculated value of the statistic and the critical value (table value).

Continued Unit - IV on F - distribution ...

F - test for significance of difference of variances:

To test for the equality of population variances F - distribution test is used.

Working Procedure:

Step - 1: Let two independent random samples of sizes n_1 and n_2 be drawn from two normal populations.

Step - 2: To test the hypothesis that the population variances σ_1^2 and σ_2^2 are equal let the null hypothesis be $H_0 : \sigma_1^2 = \sigma_2^2$.

The alternative hypothesis be $H_1 : \sigma_1^2 \neq \sigma_2^2$.

Step - 3: The sample variances that are estimates for s_1^2 and s_2^2 are given by

$$s_1^2 = \frac{\sum (x_i - \bar{x})^2}{(n_1 - 1)} \text{ and}$$

$$s_2^2 = \frac{\sum (y_i - \bar{y})^2}{(n_2 - 1)}$$

Step - 4: Test statistic $F = \frac{s_1^2}{s_2^2}$ (or) $F = \frac{s_2^2}{s_1^2}$ according as the greater value in the numerator.

Conclusion: If F - calculated value (F_{cal}) is greater than the F - tabulated value (F_{tab}) (i.e. (F_{cal}) $>$ F_{tab}) with $(n_1 - 1, n_2 - 1)$ d.f. (degrees of freedom) at 5% level of significance then we reject the null hypothesis and conclude that the variances σ_1^2 and σ_2^2 are not equal. Otherwise we accept the null hypothesis H_0 and conclude that σ_1^2 and σ_2^2 are equal.

Continued Unit - IV problems on F - distribution ...

Problem - 01: In some sample of 8 observations from a normal population, the sum of the squares of the deviations of the sample values from the sample mean is 84.4 and in another sample of 10 observations it was 102.6. Test at 5 % Los whether the population have the same variance.

Solution: Let σ_1^2 and σ_2^2 be the variances of the two normal populations from which the samples are drawn.

Null hypothesis $H_0 : \sigma_1^2 = \sigma_2^2$.

Alternative hypothesis $H_1 : \sigma_1^2 \neq \sigma_2^2$.

by data $n_1 = 8$; $n_2 = 10$

$\sum (x_i - \bar{x})^2 = 84.4$; $\sum (y_i - \bar{y})^2 = 102.6$

Therefore,

$$s_1^2 = \frac{\sum (x_i - \bar{x})^2}{(n_1 - 1)} = \frac{84.4}{(8 - 1)} \frac{84.4}{(7)} = 12.0571$$

and

$$s_2^2 = \frac{\sum (y_i - \bar{y})^2}{(n_2 - 1)} = \frac{102.6}{(10 - 1)} \frac{102.6}{(9)} = 11.4$$

$$F_{cal} = \frac{s_1^2}{s_2^2} = \frac{12.0571}{(11 - 4)} = 1.057$$

Continued Unit - IV problems on F - distribution ...

Therefore, F - calculated value = $F_{cal} = 1.057$

degrees of freedom are $v_1 = n_1 - 1 = 8 - 1 = 7$ and $v_2 = n_2 - 1 = 10 - 1 = 9$

Therefore, the tabulated value of F at 5% level for $(n_1 - 1, n_2 - 1) = (7, 9)$ d.f. is 3.29

Therefore, $F_{cal} < F_{tab}(F_{0.05}(7, 9))$

Hence, we accept null hypothesis H_0 and conclude that the population have the same variance.

Problem - 02: Pumpkin's were grown under two experimental conditions. Two random samples of 11 and 9 Pumpkin's, Show the sample s.D.'s of their weights as 0.8 and 0.5 respectively. Assuming that the weight distributions are normal, test hypothesis that true variances are equal.

Solution: Let σ_1^2 and σ_2^2 be the variances of the two normal populations from which the samples are drawn.

Null hypothesis $H_0 : \sigma_1^2 = \sigma_2^2$.

Alternative hypothesis $H_1 : \sigma_1^2 \neq \sigma_2^2$.

by data $n_1 = 11$; $n_2 = 9$; $s_1 = 0.8$; $s_2 = 0.5$

Therefore, $s_1^2 = 0.64$; $s_2^2 = 0.25$

Continued Unit - IV problems on F - distribution ...

We know that, by F - distribution for calculated value of F is

$$F_{cal} = \frac{s_1^2}{s_2^2} = \frac{0.64}{(0 - 25)} = 2.56$$

$$F_{tab} = F_{0.05}(10, 8) = 3.35$$

$\rightarrow F_{tab} = F_{0.05}(10, 8)$ is the table value of F with the d. f. (10,8) is 3.35.

Therefore, $F_{cal} < F_{tab}(F_{0.05}(7, 9))$

Hence, null hypothesis H_0 is accepted and conclude that the variances of the two populations are same. Therefore two samples have the same variance.

Problem - 03: The time taken by workers in performing a job by method I and method II is given below.

Method I	20	16	26	27	23	22	-
Method II	27	33	42	35	32	34	38

Do the data show that the variances of time distribution from population from which these samples are drawn do not differ significantly.

Solution: Let σ_1^2 and σ_2^2 be the variances of the two normal populations from which the samples are drawn.

Continued Unit - IV problems on F - distribution ...

Null hypothesis $H_0 : \sigma_1^2 = \sigma_2^2$.

Alternative hypothesis $H_1 : \sigma_1^2 \neq \sigma_2^2$.

by data $n_1 = 6$; $n_2 = 7$, $\sum x = 134$ and $\sum y = 241$

Mean values are, $\bar{x} = \frac{\sum x}{n_1} = \frac{134}{6} = 22.3$

$\bar{y} = \frac{\sum y}{n_2} = \frac{241}{7} = 34.4$

$\sum (x_i - \bar{x})^2 = 81.34$, $\sum (y_i - \bar{y})^2 = 133.72$

Therefore, $s_1^2 = \frac{\sum (x_i - \bar{x})^2}{n_1 - 1} = \frac{81.34}{5} = 16.264$

$s_2^2 = \frac{\sum (y_i - \bar{y})^2}{n_2 - 1} = \frac{133.72}{6} = 22.29$

We know that, by F - distribution for calculated value of F is

$$F_{cal} = \frac{s_2^2}{s_1^2} = \frac{22.29}{(16.264)} = 1.3699 \cong 1.37$$

The table value of F at 5% with (5,6) d.f. is

$$F_{tab} = F_{0.05}(10, 8) = 4.39$$

Therefore, $F_{cal} < F_{tab}(F_{0.05}(5, 6))$

Hence, null hypothesis H_0 is accepted and conclude that there is no significant difference between the variances of the time distribution by the workers.

Continued Unit - IV on χ^2 (Chi - Square) distribution of goodness of fit

...

This test is applied to decide whether the discrepancy between theory and experiment is significant or not.

Let the null hypothesis H_0 be that there is no significant difference between the observed values and the corresponding expected values.

The alternative hypothesis H_1 is that the above difference is significant.

Let $(O_1, O_2, O_3, \dots, O_n)$ be a set of observed frequencies and $(e_1, e_2, e_3, \dots, e_n)$ the corresponding set of expected frequencies. Then the statistic Chi - Square (χ^2) is given by

$$\chi^2 = \frac{\sum_{i=1}^n (O_i - e_i)^2}{e_i}$$
$$\chi^2 = \frac{(O_1 - e_1)^2}{e_1} + \frac{(O_2 - e_2)^2}{e_2} + \frac{(O_3 - e_3)^2}{e_3} + \dots + \frac{(O_n - e_n)^2}{e_n}$$

If the calculated value of χ^2 is greater than the tabulated value of χ^2 , at α level, then the null hypothesis H_0 is rejected otherwise the null hypothesis H_0 is accepted.

Note: No theoretical(observed) or expected frequency should be less than ten. If small frequencies occur, the difficulty is overcome by grouping two or more frequencies. Then the degrees of freedom is determined with the number of classes after regrouping.

Continued Unit - IV problems on χ^2 (Chi - Square) distribution ...

Problem - 01: The number of automobile accidents per week in a certain community are as follows: 12, 8, 20, 2, 14, 10, 15, 6, 9, 4. Are these frequencies in agreement with the belief that accident conditions were the same during this 10 week period.

Solution: The total frequency = $12 + 8 + 20 + 2 + 14 + 10 + 15 + 6 + 9 + 4 = 100$
Therefore expected frequency of accident each week = $\frac{100}{10} = 10$

Null hypothesis : H_0 : The accident conditions were same during the 10 week period.

Alternative hypothesis : H_1 : Conditions are different during the 10 week period

Therefore, the calculated value of χ^2_{cal} is

$$\chi^2_{cal} = \frac{\sum_{i=1}^{10} (O_i - e_i)^2}{e_i}$$

$$\chi^2_{cal} = \frac{(O_1 - e_1)^2}{e_1} + \frac{(O_2 - e_2)^2}{e_2} + \frac{(O_3 - e_3)^2}{e_3} + \dots + \frac{(O_{10} - e_{10})^2}{e_{10}}$$

$$\chi^2_{cal} = \frac{(12 - 10)^2}{10} + \frac{(8 - 10)^2}{10} + \frac{(20 - 10)^2}{10} + \dots + \frac{(4 - 10)^2}{10}$$

$$\chi^2_{cal} = 26.6$$

Continued Unit - IV problems on χ^2 (Chi - Square) distribution ...

Therefore, the calculated value of χ_{cal}^2 is $\chi_{cal}^2 = 26.6$

The tabulated value of χ^2 for $n = 10$ observations with $(n - 1) = 10 - 1 = 9$, d. f. at 5% is $\chi_{tab}^2 = 16.9$

Therefore, the calculated value of χ_{cal}^2 is greater than the tabulated value of χ_{tab}^2 , Since, $\chi_{cal}^2 > \chi_{tab}^2$, hence the null hypothesis H_0 is rejected and conclude that the accident conditions were not the same during the 10 week period.

Problem - 02: A die is thrown 264 times with the following results. Show that the die is biased . Given that $\chi_{0.05}^2 = 11.07$ for 5 d.f.

Number appeared on the die	1	2	3	4	5	6
frequency	40	32	28	58	54	52

Solution: Let the null hypothesis H_0 : The die is unbiased; alternative hypothesis H_1 : die is biased.

The expected frequency of each of the numbers (1, 2, 3, 4, 5, 6) is $\frac{264}{6} = 44$

Continued Unit - IV problems on χ^2 (Chi - Square) distribution ...

Therefore, the calculated value of χ_{cal}^2 is

$$\chi_{cal}^2 = \frac{\sum_{i=1}^6 (O_i - e_i)^2}{e_i}$$
$$\chi_{cal}^2 = \frac{(O_1 - e_1)^2}{e_1} + \frac{(O_2 - e_2)^2}{e_2} + \frac{(O_3 - e_3)^2}{e_3} + \dots + \frac{(O_6 - e_6)^2}{e_6}$$
$$\chi_{cal}^2 = \frac{(40 - 44)^2}{44} + \frac{(32 - 44)^2}{44} + \frac{(28 - 44)^2}{44} + \dots + \frac{(52 - 44)^2}{44}$$
$$\chi_{cal}^2 = 17.6362$$

The tabulated value of $\chi^2 - tab$ for $n = 6$ observations with $(n - 1) = 6 - 1 = 5$, d. f. at 5% is $\chi_{tab}^2 = 11.07$

Therefore, the calculated value of χ_{cal}^2 is greater than the tabulated value of χ_{tab}^2 , Since, $\chi_{cal}^2 > \chi_{tab}^2$ and hence the null hypothesis H_0 is rejected and conclude that the die is biased.

Problem - 03: Fit a Poisson distribution to the following data and for its goodness of fit at 5% level of significance.

Continued Unit - IV problems on χ^2 (Chi - Square) distribution ...

x	0	1	2	3	4
f	419	352	154	56	19

Solution:

x	f	f x
0	419	0
1	352	352
2	154	308
3	56	168
4	19	76
	$N = \sum f = 1000$	$\sum fx = 904$

Therefore, the mean $= \bar{x} = \frac{\sum fx}{\sum f} = \frac{904}{1000} = 0.904$

The theoretical distribution is given by $= NXp(x) = 1000X(\frac{e^{-m}m^x}{x!})$, $\rightarrow m = 0.904$

Theoretical frequencies are given by $= f(x) = 1000X\frac{e^{-0.904}(0.904)^x}{x!}$, for $x = 0, 1, 2, 3, 4$, we get

x	0	1	2	3	4	Total
f	404 (406.9)	366	165.4	44.8	11.3 (12.6)	997.4

Continued Unit - IV problems on χ^2 (Chi - Square) distribution ...

As sum is not equal to 1000, the first and last theoretical frequencies 406.2 and 12.6 are taken as 404.9 and 11.3 as shown in the brackets.

Therefore, the calculated value of χ_{cal}^2 is

$$\chi_{cal}^2 = \frac{\sum_{i=0}^4 (O_i - e_i)^2}{e_i}$$

$$\chi_{cal}^2 = \frac{(O_0 - e_0)^2}{e_0} + \frac{(O_1 - e_1)^2}{e_1} + \frac{(O_2 - e_2)^2}{e_2} + \dots + \frac{(O_4 - e_4)^2}{e_4}$$

$$\chi_{cal}^2 = \frac{(404.9 - 0.904)^2}{0.904} + \frac{(366 - 0.904)^2}{0.904} + \frac{(165.4 - 0.904)^2}{0.904} + \dots + \frac{(11.3 - 0.904)^2}{0.904}$$

$$\chi_{cal}^2 = 5.748$$

The tabulated value of $\chi^2 - tab$ for $n = 5$ observations with $(n - 2) = 5 - 2 = 3$, d. f. at 5% is $\chi_{tab}^2 = 7.82$

Therefore, the calculated value of χ_{cal}^2 is less than the tabulated value of χ_{tab}^2 ,

Since, $\chi_{cal}^2 < \chi_{tab}^2$ and hence this agreement between the fact and theory is good, so the Poisson distribution can be fitted to the data.

Continued Unit - IV problems on Test of significance for single mean (Large Samples) ...

Problem - 01: A sample of 64 students have a mean weight of 70kgs. Can this be regarded as a sample from a population with mean weight 56kgs and standard deviation 25kgs.

Solution: By data the mean of the samples = $\bar{x}$ 70kgs,

the mean of the population = $\mu = 56kgs$

the standard deviation of the population = $\sigma = 25kgs$

the sample size = $n = 64$

Let the null hypothesis : $H_0 : \mu = \mu_0 = 56$

That is, A sample of 64 students with mean weight of 70kgs can be regarded as a sample from a population with mean weight 56kgs and standard deviation 25kgs.

Alternative hypothesis : $H_1 : (\mu > 56)$ (OR) $(\mu < 56)$

that is, sample cannot be regarded as one coming from the population

Level of significance is 5% (when the LOS is not given by default take 5%)

Test statistic : z

$$z = \frac{(\bar{x} - \mu)}{\left(\frac{\sigma}{\sqrt{n}}\right)} = \frac{(70 - 56)}{\left(\frac{25}{\sqrt{64}}\right)} = 4.48$$

Therefore, the calculated value of test statistic (z) = is $z_{cal} = 4.48$

the tabulated value of test statistic(z) at 5% LOS is $z_{tab} = z_{\alpha} = z_{0.05} = 1.645$

Clearly, $z_{cal} > z_{tab}$ and hence the null hypothesis is rejected.

Continued Unit - IV problems on Test of significance for single mean (Large Samples) ...

Problem - 02: An oceanographer wants to check whether the depth of the ocean in a certain region is 57.4 fathoms, as had previously been recorded. What can be conclude at the 5% level of significance, if readings taken at 40 random locations in the given region yielded a mean of 59.1 fathoms with a standard deviation of 5.2 fathoms.

Solution: Given $n = 40$, $\bar{x} = 59.1$ and $\sigma = 5.2$

Let us assume that

H_0 : Null hypothesis : $\mu = 57.4$

H_1 : Alternative hypothesis : $\mu \neq 57.4$

Level of significance = 0.05

The calculated Test statistic : $t_{cal} = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{59.1 - 57.4}{\frac{5.2}{\sqrt{40}}} = 2.067$

The table value is $t_{tab} = t_{\alpha} = t_{0.05} = 1.96$

Therefore, $t_{cal} > t_{tab}$ and hence the Null hypothesis H_0 is rejected at 5% level of significance.

Problem - 03: The means of two large samples of sizes 1000 and 2000 members are 67.5 inches and 68.0 inches respectively. Can the samples be regarded as drawn from the same population of standard deviation 2.5 inches.

Solution: Let μ_1 and μ_2 be the means of the two populations. Given $n_1 = 1000$, $n_2 = 2000$, $\bar{x}_1 = 67.5$, $\bar{x}_2 = 68$ inches and $\sigma = 2.5$ inches

Continued Unit - IV problems on Test of significance for single mean (Large Samples) ...

Let us assume that

H_0 : Null hypothesis : $\mu_1 = \mu_2$ (Sample drawn from the sample proportion)

H_1 : Alternative hypothesis : $\mu_1 \neq \mu_2$

Level of significance = 0.05

The calculated Test statistic : $t_{cal} = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2}}} = -5.16$

The table value is $t_{tab} = t_{\alpha} = t_{0.05} = 1.96$

Therefore, $|t_{cal}| = 5.16 > t_{tab} = 1.96$ and hence the Null hypothesis H_0 is rejected at 5% level of significance.

Problem - 04: A sample of 100 bulbs produced by a company A showed a mean life of 1190 hours and a standard deviation of 90 hours. Also a sample of 75 bulbs produced by a company B showed a mean life of 1230 hours and a standard deviation of 120 hours.

Continued Unit - IV problems on Test of significance for single mean (Large Samples) ...

Problem - 04: Samples of students were drawn from two universities and from their weights in kilograms, mean and S. D.'s are calculated and shown below. Make a large sample test to test the significance of the difference between the means.

	Mean	S. D.	Size of the sample
University A	55	10	400
University B	57	15	100

Solution: Given $\bar{x}_1 = 55$, $\bar{x}_2 = 57$, $n_1 = 400$, $n_2 = 100$, $s_1 = 10$ and $s_2 = 15$ (Sample S. D.'s). Let us assume that

Null hypothesis: $H_0 : \bar{x}_1 = \bar{x}_2$

Alternative hypothesis: $H_1 : \bar{x}_1 \neq \bar{x}_2$

at the 5% LOS, We have, the Test statistic

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = -1.26$$

$$|z_{cal}| = |-1.26| = 1.26$$

The tabulated value at the 5% LOS is $z_{tab}(0.05) = 1.96$. Therefore, $z_{cal} < z_{tab}$. hence we accept the null hypothesis at 5% LOS and conclude that there is no significant difference between the means.

Method - I: Let $\bar{x}$ = mean of a sample, n = size of the sample, s = S. D.'s of the sample and μ = mean of the population. Then the statistic of the students t - test is defined as

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

Sample variance with the sample size n is

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

The variance of the sample s^2 is called the unbiased estimate of population variance σ^2 .

Assumptions of the Students t - test:

- ① Sample size $n > 30$.
- ② The parent population from which sample is drawn is normal.
- ③ The population S. D. is unknown.
- ④ The sample observations are independent (i. e. sample is random)

Continued Unit - IV problems on Students t - test OR Students t - distribution ...

Problem - 01: A mechanist is making engine parts with axle diameter of 0.700 inch. A random sample of 10 parts shows a mean diameter of 0.742 inch with S. D. of 0.040 inch. Compute the statistic you would use to test whether the work is meeting the specification at 5% level of significance.

Solution: Given sample size $n = 10$ (small sample), $\bar{x} = 0.742$, $\mu = 0.700$ and S.D. $s = 0.040$

Null hypothesis = $H_0 : \mu = 0.700$ The product is confirming to specification

Alternative hypothesis = $H_1 : \mu \neq 0.700$

We have, the statistic of the student t - test

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{0.742 - 0.700}{\frac{0.040}{\sqrt{10}}} = 3.32$$

Therefore, the calculated value is $t_{cal} = 3.32$

The tabulated value with $(n - 1) = (10 - 1) = 9$ d. f. at 5% LOS is $t_{tab} = 2.262$ and $t_{cal} > t_{tab}$. Therefore, the null hypothesis H_0 is rejected and hence we conclude that the product is not meeting the specification.

Continued Unit - IV problems on Students t - test OR Students t - distribution ...

Problem - 02: A certain stimulus administered to each of the 12 patients results in the following change in blood pressure 5, 2, 8, -1, 3, 0, 6, -2, 1, 5, 0, 4. Can it be concluded that the stimulus will increase the blood pressure. Given $t_{0.05}$ with 11 d. f. is 2.201.

Solution: We have to find the mean $= \bar{x} = \frac{\sum x}{n} = \frac{31}{12} = 2.5833$
 $n = 12$ and the sample variance is

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1} = \frac{(5 - 2.5833)^2}{11} + \frac{(2 - 2.5833)^2}{11} + \frac{(8 - 2.5833)^2}{11} + \dots + \frac{(4 - 2.5833)^2}{11} = 9.538$$

Sample S. D. $= s = 3.0883$ and we shall frame the hypothesis

Null hypothesis : $H_0 = \mu = 0$ Stimulus administration is not accompanied with increase in B. P. (The average change in B. P.)

Alternative hypothesis : $H_1 = \mu \neq 0$ Stimulus administration is not accompanied with increase in B. P.

The calculated value of student t - test is

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{2.5833 - 0}{\frac{3.088}{\sqrt{12}}} = 2.8979 \approx 2.9$$

$t_{cal} = 2.9$ The tabulated value with $(12 - 1) = 11$ d.f. at 5% is $t_{tab} = 2.201$.

Continued Unit - IV problems on Students t - test OR Students t - distribution ...

Therefore, $t_{cal} > t_{tab}$ and the null hypothesis H_0 is rejected at 5% LOS. Hence we conclude that the stimulus is accompanied with increase in B. P.

Problem - 03: A random sample of 10 boys had the following I.Q.'s 70, 120, 110, 101, 88, 83, 95, 98, 107 and 100. Do these data support the assumption of a population mean I_Q's of 100,

Solution: We have to find the mean $= \bar{x} = \frac{\sum x}{n} = \frac{972}{10} = 97.2$
 $n = 12$ and the sample variance is

$$s^2 = \frac{\sum (x_i - \bar{x})}{n - 1} = \frac{(70 - 97.2)}{9} + \frac{(120 - 97.2)}{9} + \frac{(110 - 97.2)}{9} + \dots + \frac{(100 - 97.2)}{9} = 203.93$$

Sample S. D. $= s = 14.27$ and we shall frame the hypothesis

Null hypothesis : $H_0 = \mu = 100$ The data support the assumption at a population mean I. Q.'s of 100 in the population.

Alternative hypothesis : $H_1 = \mu \neq 100$ The data is not support the assumption at a population mean I. Q.'s of 100 in the population.

The calculated value of student t - test is

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{97.2 - 100}{\frac{14.7}{\sqrt{10}}} = -0.6023$$

Continued Unit - IV problems on Students t - test OR Students t - distribution ...

$t_{cal} = |t_{cal}| = |-0.6023| = 0.6023$ The tabulated value with $(10 - 1) = 9$ d.f. at 5% is $t_{tab} = 2.262$.

Therefore, $t_{cal} < t_{tab}$ and the null hypothesis H_0 is accepted at 5% LOS. Hence we conclude that the data support the assumption of mean I. Q.'s of 100 in the population.

Method - II - Student t - test for difference of means:

Let $\bar{x}_1$ and $\bar{x}_2$ are the means of two independent samples of sizes n_1 and n_2 drawn from two normal populations having means μ_1 and μ_2 respectively. To test whether the population means are equal (i.e. whether the difference means $\mu_1 - \mu_2$ is significant). We shall frame the hypothesis

Null hypothesis : $H_0\mu_1 = \mu_2$: The population mean is equal.

Alternative hypothesis : $H_1\mu_1 \neq \mu_2$: The population mean is not equal.

If S. D.'s $\sigma_1 = \sigma_2 = \sigma$, then an unbiased estimate s^2 of the common variance σ^2 is given by

$$s^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{(n_1 + n_2 - 2)}$$

where, s_1^2 and s_2^2 are the sample variances. Also, the standard error is S. E. of $(\bar{x}_1 - \bar{x}_2) = s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$, where, $s = \frac{n_1 s_1^2 + n_2 s_2^2}{(n_1 + n_2 - 2)}$.

Assuming that the null hypothesis H_0 is true, the student test statistic t is given by

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Continued Unit - IV problems on Students t - test OR Students t - distribution ...

follows t - distribution with $(n_1 + n_2 - 2)$ d. f.

Here,

$$s^2 = \frac{1}{(n_1 + n_2 - 2)} \left[\sum (x_i \bar{x}_1)^2 + \sum (x_j \bar{x}_1)^2 \right]$$

Problem - 01: Samples of two types of electric bulbs were tested for length of life and following data were obtained

Type - I	Type - II
Sample number = $n_1 = 8$, sample mean = $\bar{x}_1 = 1234$ hours, sample S. D. = $s_1 = 36$ hours	Sample number = $n_2 = 7$, sample mean = $\bar{x}_2 = 1036$ hours, sample S. D. = $s_2 = 40$ hours

is the difference in the means sufficient to warrant the type - I is superior to type - II regarding length of life.

Solution: Since the sample sizes are small and the S.D.'s σ_1, σ_2 are not known.

Let μ_1 and μ_2 be the population means. We shall frame the hypothesis

Null hypothesis : $H_0 \mu_1 = \mu_2$: The two types I and II of electric bulbs are identical.

Alternative hypothesis : $H_1 \mu_1 \neq \mu_2$: The two types I and II of electric bulbs are not identical.

Continued Unit - IV problems on Students t - test OR Students t - distribution ...

We have, the calculated value is

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

where

$$s^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{(n_1 + n_2 - 2)} = \frac{8(36)^2 + 7(40)^2}{8 + 7 - 2} = 1659.08$$

$$t_{cal} = \frac{1234 - 1036}{1659.08 \sqrt{\frac{1}{8} + \frac{1}{7}}} = 9.39$$

$t_{cal} = 9.39$ The tabulated value with $(8 + 7 - 2) = 13$ at 5% LOS $t_{tab} = 2.160$.

Therefore, $t_{cal} > t_{tab}$ and the null hypothesis H_0 is rejected. Hence we conclude that the two types I and II of electric bulbs are not identical.

Continued Unit - IV problems on Students t - test OR Students t - distribution ...